Data management lab

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Instructions

This lab is dedicated to the analysis of a data set described below. Your work must be submitted as a R project in a zip file. You must:

- create a R project on your computer and do all your work in this project;
- unzip the zip file obtained on moodle within the R project (all your data files must be in as sub-directory of your project, see question 1);
- create a quarto document using **your first name** as the new name;
- at the end of the session prepare to upload a zip file on moodle with at least:
  - the R project file (ending with `.Rproj`);
  - the data files (csv format);
  - the quarto document;
  - the result of rendering your document to html.
- note that
  - answers can be written in English or French;
  - all graphical representations must be done with ggplot2. The use of base R plot using `plot` function and variants is not authorised;
  - all calculations must be done using dplyr and tidyr;
  - you are authorised to use any documentation during the test, both with the help files and using web sites and other online sources;
  - you may also use AI tools to get help for the code, but not for comments and analysis. Note that the code produce by those tools is sometimes clunky and shows an inconsistent style, with redundencies and other limitations. This may impact your grade. Note also that AI tools will probably suggest the use of functions that are not part of dplyr or tidyr, possibly loops and other constructions not seen during the course. Anything done like this will be considered incorrect, even if the results are the expected ones.

By uploading your submission you acknowledge that it contains the result of your personal work.

Individual instructions

The instructions contained in this document are student specific. Your data sets are unique to you and parts of the instructions depend on the data sets. In particular, most names are

unique.

1 Planet data set analysis

Data loading

You data set should load with no difficulty using `vroom` or `read_csv`, but you are expected to check the validity of the reading procedure: take care of any error or warning, and make sure you have loaded the expected number of variables (with proper data types).

You main data set is contained in the file "Planet_list.csv".

Question 1

Create a data folder to store the files. Copy or move the data set files in this directory.

The main data set contains descriptions of 866 imaginary planets from a science fiction universe. Each planet is described by a collection of 8 variables:

- "`relative mass (earth)`": mass of the planet, expressed in mass of the Earth unit
- "`relative radius`": radius of the planet, expressed in radius of the Earth unit
- "`DENSITY (EARTH)`": density of the planet, expressed in density of the Earth unit
- "`P_NAME`": name of the planet
- "`Planet_number`": a unique key for each planet
- "`Star_idx`": the key of the star of the planet
- "`Terrain`": types of terrain that can be found on the planet
- "`Type_climate`": types of climate that can be found on the planet

The file contains some missing data encoded by the following value: NA. This should be recognized automatically by `vroom` or `read_csv`.

Missing values

To test if a value is missing, use the `is.na` function.

The two key variables enable identification of the planets (each value is unique in the planet data set) and of the stars. Note that a star can have multiple planets in the data set and thus that the corresponding key is not unique in the planet data set (see question 7).

Question 2

Add code to your quarto file to load the main data set. To verify the validity of the loading process, check that the number of planets is equal to the one specified above and that you get the correct number of variables. Verify also that all the numerical variables are recognised as numerical variables. Report your findings in the document, for instance in a table or a list.

Question 3

Represent graphically the density of a planet as a function its radius. Make sure to get rid of any warning or error!

Question 4

Compute summary statistics of the mass of a planet (mean, median, standard deviation and interquartile range) and include them in a table in the rendered document. You may want to use `tidyr::pivot_longer` to improve the presentation of the result.

Question 5

Compute the percentage of planets that have a radius larger than Earth radius.

Question 6

Report in a **table** the most extreme planets with respect to the numerical variables. This should include, for instance, the densest and least dense planets. You can report a single planet per characteristics. For instance, if multiple planets are the heaviest, you may show one of them only.

All names must be computed by the report, not entered as fixed quantities in the text. The use of `dplyr::slice_min` and related functions is recommended.

Question 7

Show graphically the distribution of the number of planets per star, i.e. how many stars have one planet, two planets, etc. You have first to compute the number of planet per star key.

2 Star data

The planet data set is complemented by a star data contained in the file "DATA-STARS.csv". It contains descriptions of 561 imaginary stars from a science fiction universe. Each star is described by a collection of 8 variables:

- "`STAR`": the name of the star
- "`DISTANCE (LIGHTYEARS)`": distance in light years from the Sun
- "`Type Of Star`": stellar type of the star
- "`Star_idx`": a unique key for each star
- "`Star Temperature`": effective temperature in Kelvin
- "`Star Radius`": radius of the star, expressed in solar radii
- "`relative mass`": mass of the star, expressed in solar masses
- "`luminosity`": luminosity of the star, relative to the Sun's luminosity

Question 8

Add code to load the data set.

The stars are classified into different stellar types. A stellar type is made of three elements:

- a letter according to the *Harvard system*, which gives essentially the temperature *class* of the star, see the `star_types.csv` file;
- a digit from 0 to 9 that refines this temperature into a *magnitude* (0 for maximal temperature, 9 for minimal);
- a short text according to the *Yerkes spectral classification* which gives the *luminosity* class of the star (it is also related to its surface gravity), see the `Yerkes_luminosity_classes.csv` file.

In the star data set, the three components are merged in the `Type Of Star` variable using the separator `|`.

Question 9

Split the stellar type into three variables, one per component. This can be done using the `tidyr::separate_wider_delim` function.

Question 10

Compute the average mass of the stars per temperature class, that is the average mass of stars of class 0, B, etc. Include the results as a table in your document.

3 Combined analysis

> **Combining two data frames**
>
> Some of the following questions can only be answered by combining the planet data frame with the star data frame, using the `Star_idx` variable. Provided the data frames are `planets` and `stars` this can be done as follows:
>
> ```
> ```{r}
> planet_star <- inner_join(planets, stars, by = join_by(Star_idx))
> ```
> ```
>
> The `planet_star` data frame will contain all the planets described by their original variables, complemented by the variables describing the star of each planet. Notice than some stars may be present in the star data set but without any corresponding planets. They will not appear in `planet_star`.
> This approach can be used to combine any pair of data frame that share a common key column. For instance, one can compute some quantities related to stars, building a new data frame with a `Star_idx` variable. This data frame can then be combined with the `stars` data using a call to `inner_join`. This applies to any data frame, in particular to the planets in this lab.

Question 11

Compute the number of stars described in the star data set that do not have a least one corresponding planet. You may do that using the `dplyr::distinct` function, for instance (there are many other ways to answer to this question).

Question 12

Compute for each star the average of the radius of its planets and plot this quantity as a function of the radius of the star (make sure the graphical representation is readable). To do that, you may need to use `inner_join` between two data frames that share a common column.

4 Planet characteristics

The terrain and climate variables are *multivalued*: a given planet can have multiple climates and multiple types of terrain. In the main data set, all values of e.g. terrain are kept in a single variable, `Terrain`. The different values are separated by the "|" symbol.

Question 13

Using `dplyr::select` and `tidyr::separate_longer_delim`, extract the terrain types and the climate types into two different data frames containing only two columns: the planet identifier (`Planet_number`) and the quantity of interest (terrain type or climate type). This last column must contain only a single value: there will be multiple rows per planet in general.

Question 14

Represent graphically the distribution of the number of different terrain types per planet.

Question 15

Represent graphically the distribution of the mass of the planets conditioned by the climates. In other words, we want to see the distribution of the mass of the planets that have some *arid* regions, some *humid* regions, etc. Notice that each planet will be accounted for multiple times owing to the multivalued nature of the terrain and climate variables. This is not a problem! You may need to use again `dplyr::select` and `tidyr::separate_longer_delim` to simplify your task.

Question 16

Compute for each star the total number of distinct climates seen on its surrounding planets.